Poorni Pathirage

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Third-year Computer Engineering undergraduate at the University of Sri Jayewardenepura with a research focus on deep neural networks, currently investigating structural pruning and model compression to make neural networks smaller and faster for edge devices. Experienced in machine learning, data science and NLP, from building end-to-end data pipelines to evaluating models across CNN, Transformer, RNN and graph architectures. Also skilled in cloud computing on AWS and full-stack development, with a track record of turning research ideas into practical, well-tested systems.

Education

University of Sri Jayewardenepura *Feb 2024 – Present*
BSc Engineering (Hons) in Computer Engineering

- GPA: 3.38/4.00
- Minor: Data Management
- Core Modules: Data Structures and Algorithms, Object-Oriented Programming (OOP), Natural Language Processing, Data Mining, Cloud Computing and Applications

G.C.E. Advanced Level *2020 – 2022*
Physical Science Stream, Visakha Vidyalaya

- 3A passes (Z-score 1.9584)

Projects

Structural Pruning of Neural Networks: Systematic Evaluation and Extension of DepGraph *Oct 2026 – Present*
Individual Research Project – Supervisor: Sanka Mohottala
External Supervisors: Mr. Asiri Gawesha (SLIIT), Dr. Dinuka Sahabandu (University of Washington)

- Researching **structural pruning**, a model compression technique that removes whole channels, neurons and attention heads to make deep neural networks smaller and faster for edge devices.
- Studying **DepGraph (CVPR 2023)** and the open-source Torch-Pruning library to evaluate which architectures it supports, including CNN, RNN/LSTM, Transformer and graph neural networks.
- Developing an automated test harness in PyTorch that records whether pruning succeeds, crashes or silently produces incorrect outputs.
- Aiming to add pruning support for under-supported layers (such as LSTM/GRU) and **contribute it to the open-source Torch-Pruning library**, and to compare pruning methods fairly on accuracy, model size and inference latency.
- Reviewing related research on pruning and edge inference, including magnitude-based pruning for deployment on mobile devices.
- **Tech:** Python, PyTorch, Torch-Pruning, torchvision, Hugging Face Transformers, PyTorch Geometric

End-to-End E-Commerce ML Pipeline *Jul 2026* GitHub

- Built a complete ML pipeline on ∼**780,000** real e-commerce transactions (UCI Online Retail II), from data cleaning to an interactive dashboard.
- Segmented customers via **RFM analysis** and clustering, and built a churn prediction model with a time-based train/test split.
- Ran Apriori market-basket analysis and a PageRank co-purchase network to surface high-influence products, deployed via Streamlit.
- **Tech:** Python, pandas, scikit-learn, mlxtend, NetworkX, Streamlit

BedSense – Smart Bedsore Detection System *Jan 2026 – Present*

- Conceived the initial project idea and contributed to the development of a sensor-based healthcare monitoring prototype for detecting pressure-related risks in bedridden patients.

- Designed the software architecture, integrated the database, and developed a user-friendly software interface to support real-time patient monitoring.
- Collaborated with a multidisciplinary team to integrate software with embedded hardware and IoT communication workflows.
- **Selected among the Top 6 finalist teams (from 16 national finalists) at Hult Prize Sri Lanka Nationals 2026.**
- **Tech:** MQTT, SQLite, Node.js, React, ESP32, IoT

Awards & Honors

View on LinkedIn

- **Finalist – Hult Prize Sri Lanka Nationals 2026**, APIIT, in partnership with the Ministry of Digital Economy – national leg of the Hult Prize, a **globally recognized** student social entrepreneurship competition with a shot at **$1M** in seed funding; competed among 10+ student-led startups for a place at the Nationals and were **selected among the top 6**. (*BedSense*)
- **Finalist – SLIoT Challenge 2026**, University of Moratuwa in partnership with SLT-Mobitel & the Institution of Engineers Sri Lanka (IESL) – **Sri Lanka's largest** national IoT/AIoT competition, themed around Industry 5.0 (*BedSense*)
- **1st Runner-Up – Brainstorm 2026**, IEEE Engineering in Medicine & Biology Society (EMBS) Student Chapter, University of Moratuwa – national healthcare and biomedical engineering innovation competition (*BedSense*)
- **Finalist – Mobitel CodeBlast Hackathon 2025**, SLT-Mobitel & hSenid Mobile – national-level hackathon for app development on the mSpace platform, showcasing innovation and next-gen tech talent
- **Champions** – VERTEX 2025 Innovation Challenge, IEEE Robotics & Automation Society (RAS) Student Chapter, Informatics Institute of Technology – debut national IoT and robotics innovation challenge
- Winner – IdeaPulse 2.0 Inter-Faculty Idea Pitching Competition 2026

Key Skills

Languages: C++, Python, Java, English (Professional), Sinhala (Native)

Deep Learning & Model Compression: PyTorch, Torch-Pruning, torchvision, Hugging Face Transformers, PyTorch Geometric, Structural Pruning, CNNs, Transformers, Graph Neural Networks

Data Science / ML: pandas, scikit-learn, mlxtend, NetworkX, NLP, Customer Segmentation, Market-Basket Analysis

Embedded / Cloud: AWS Cloud Infrastructure, Embedded Systems, Sensor Integration, MQTT / IoT

Tools & Software: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Streamlit, Figma, Canva, Flutter

Certifications

- AWS Academy Graduate – Cloud Foundations, *AWS Academy* *Jul 2026* Badge
  Learned core AWS services, cloud architecture design, cloud security and Identity and Access Management (IAM).
- SQL (Basic), *HackerRank*
- Java Intermediate, *Sololearn*
- Introduction to Java, *Sololearn*
- Introduction to Python, *Sololearn*
- Introduction to C++, *Sololearn*

References

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